BITS Pilani

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Artificial and Computational Intelligence

AIMLCZG557

ACI Team

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AIMLCZG557 – CS#9

Knowledge Representation using Logics

Agenda for CS #9

1. Logic Representation of a sample agent
2. TT-Entail for inference from truth table
3. Propositional theorem proving – Proof by resolution
4. Propositional theorem proving – Proof by Contradiction
5. PL-Resolution using CNF Conversion and unit resolution
6. DPLL Algorithm
7. Introduction to Predicate Logic
8. Predicate Logic resolution using Forward & Backward chaining
9. Exercises

Course Progress

M1 Introduction to AI

M2 Problem Solving Agent using Search

M3 Game Playing

M4 Knowledge Representation using Logics

M5 Probabilistic Representation and Reasoning

M6 Reasoning over time

M7 Ethics in AI

Knowledge-based Agents

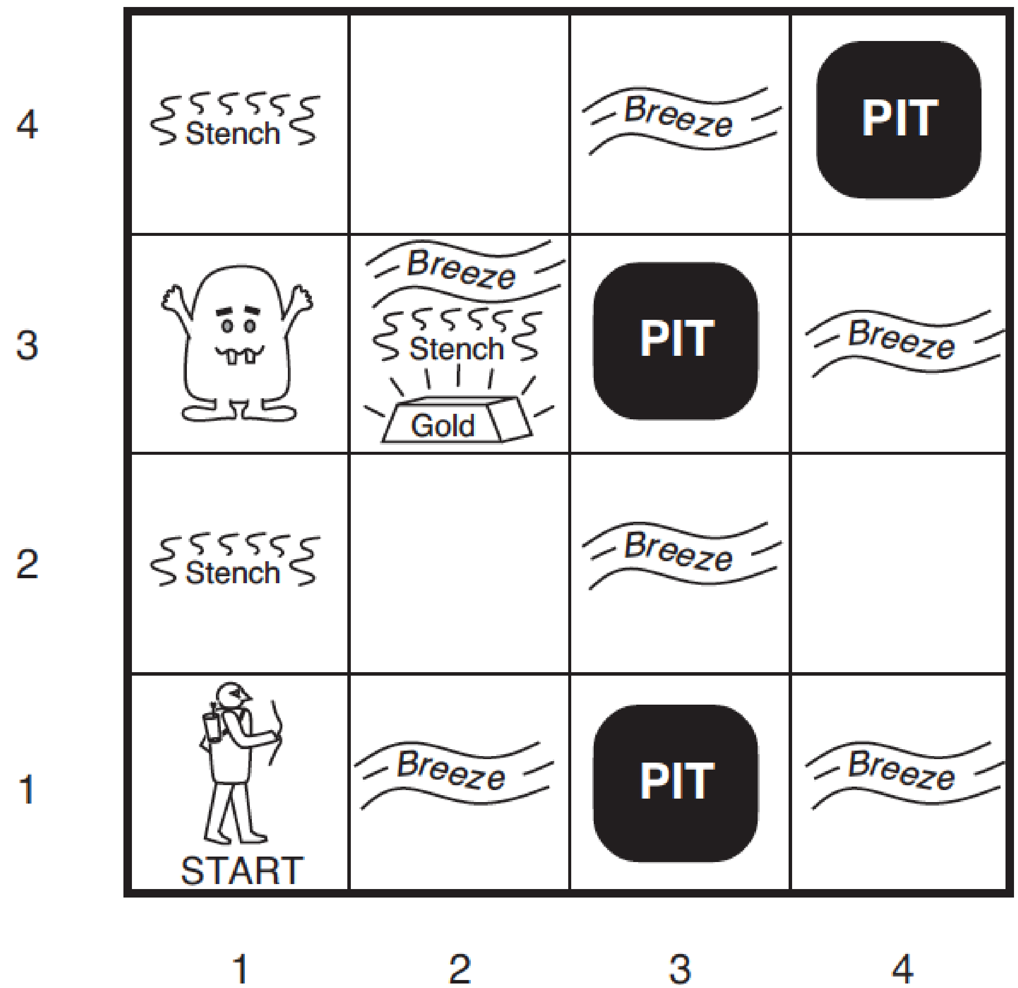
- Solving problems by
  - Representing the knowledge in state-space
  - Reasoning about the solution in logical steps
- Central Component: Knowledge Base (KB)
- KB – set of sentences, not the English sentences
- Sentence
  - representation of knowledge in a language called Knowledge representation language
  - Represents an axiom, when the sentence is taken as given without being derived from other sentences
  - TELL operation: Add new sentences to the knowledge base
  - ASK operation: Query what is known

Knowledge-based Agents

- Each time a knowledge-based agent is called, it does three things
  - TELL – the Knowledge Base about the percepts (inputs)
  - ASK – the Knowledge Base what action it should perform.
    - Extensive reasoning about the outcomes of possible action and current state
  - TELL – the Knowledge Base about selected action (to update) and agent executes the action

Example: Knowledge-based Agents

Wumpus World



Wumpus World

- Defining the task environment (PEAS)
- Performance Measure:
  - +1000 for climbing out with gold,
  - -1000 for falling into a pit or being eaten by Wumpus,
  - -1 for each action taken and
  - -10 for using an arrow
- Environment: 4x4 grid of rooms. Always starts at [1, 1] facing right. The location of Wumpus and Gold are random. Agent dies if entered a pit or live Wumpus.

Wumpus World

- Actuators:
  - Forward,
  - TurnLeft by 90,
  - TurnRight by 90,
  - Grab – pick gold if present,
  - Shoot – fire an arrow, it either hits a wall or kills wumpus. Agent has only one arrow.
  - Climb – Used to climb out of cave, only from [1, 1]

Wumpus World

- Sensors: The agent has five sensors
  - Stench: In all adjacent (but not diagonal) squares of Wumpus
  - Breeze: In all adjacent (but not diagonal) squares of a pit
  - Glitter: In the square where gold is
  - Bump: If agent walks into a wall
  - Scream: When Wumpus is killed, it can be perceived everywhere

- Percept Format: [Stench?, Breeze?, Glitter?, Bump?, Scream?]
- E.g., [Stench, Breeze, None, None, None]

Wumpus world Characterization

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- Fully Observable No – only local perception
- Deterministic Yes – outcomes exactly specified
- Episodic No – sequential at the level of actions
- Static Yes – Wumpus and Pits do not move
- Discrete Yes
- Single-agent? Yes – Wumpus is essentially a natural feature

- **Goal:** Discover the location of Wumpus, Pits and Gold so that the agent can navigate to Gold without falling prey to Wumpus or Pits
- This would be agent's knowledge for transition
- Requires a configuration of the environment to do logical reasoning

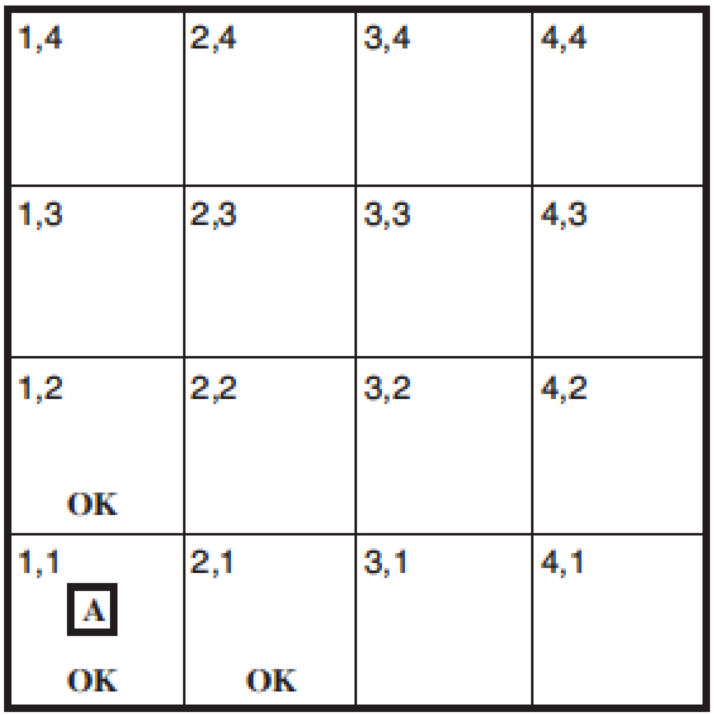
Wumpus World

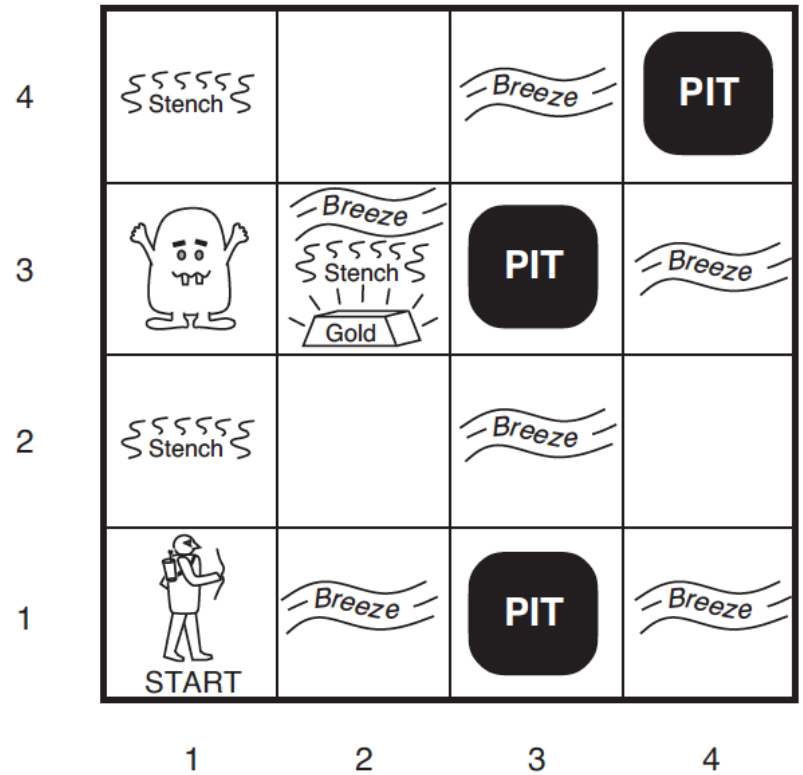
Symbolic Notation

- A = Agent
- B = Breeze
- G = Glitter, Gold
- OK = Safe Square
- P = Pit
- S = Stench
- V = Visited
- W = Wumpus

Wumpus World

Percept 1: [None, None, None, None, None]





Action: Forward

Wumpus World

Percept 1: [None, None, None, None, None]

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 $\boxed{A}$ OK	2,1 OK	3,1	4,1

Action: Forward

Wumpus World

Percept 2: [None, Breeze, None, None, None]

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 $\boxed{A}$ B OK	3,1 P?	4,1

Action: Move to [1, 2] (the other non-visited OK state)

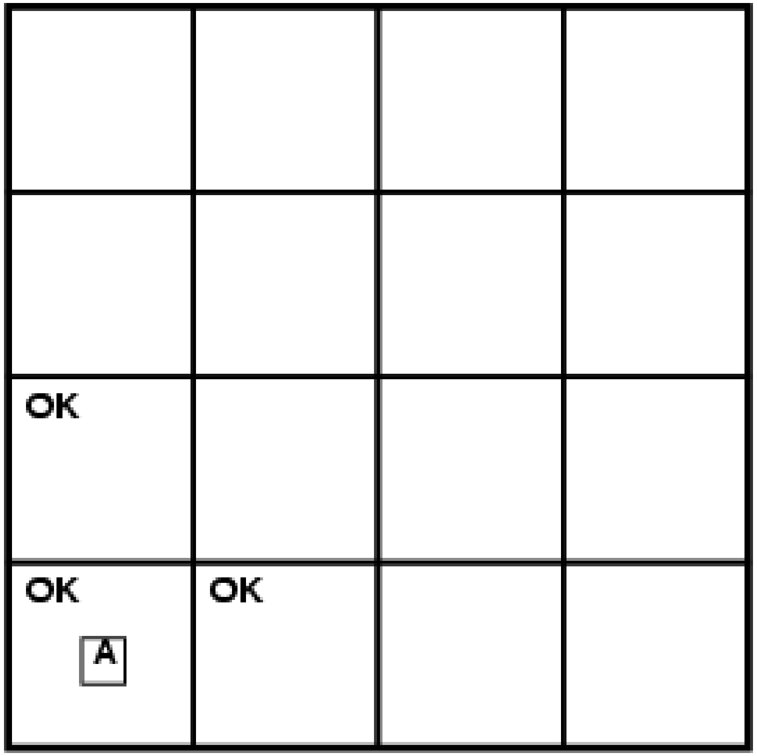
Wumpus World

Percept 3: [Stench, None, None, None, None]

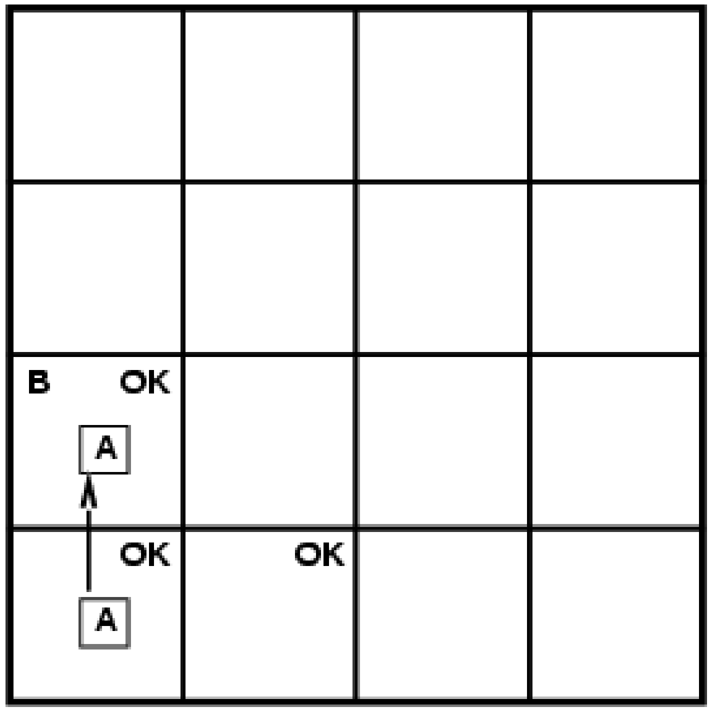
1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 $\boxed{A}$ S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

Action: Move to [2, 2]

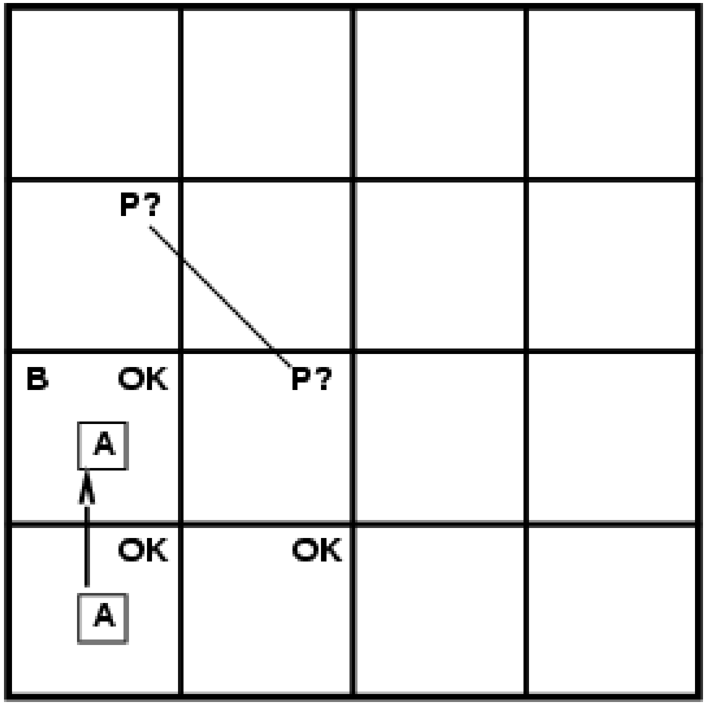
Another Wumpus World Game



Another Wumpus World Game



Another Wumpus World Game



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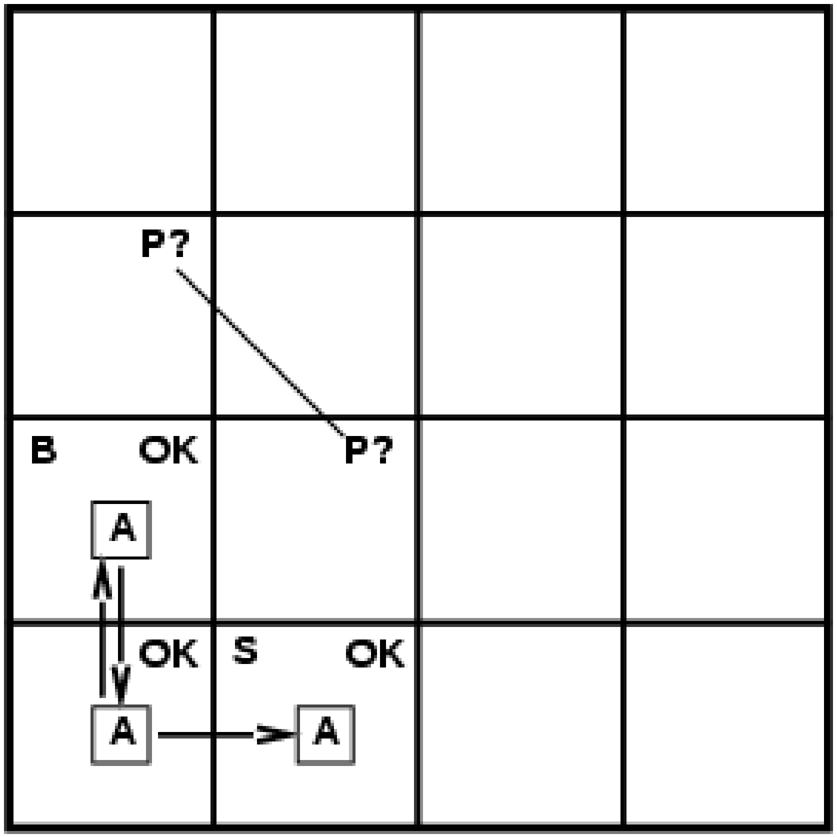

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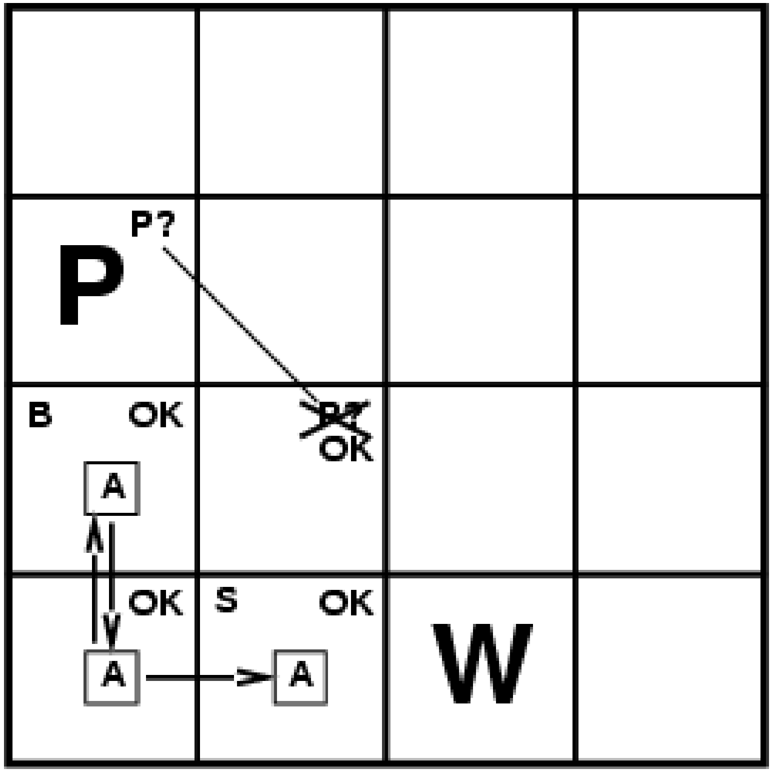
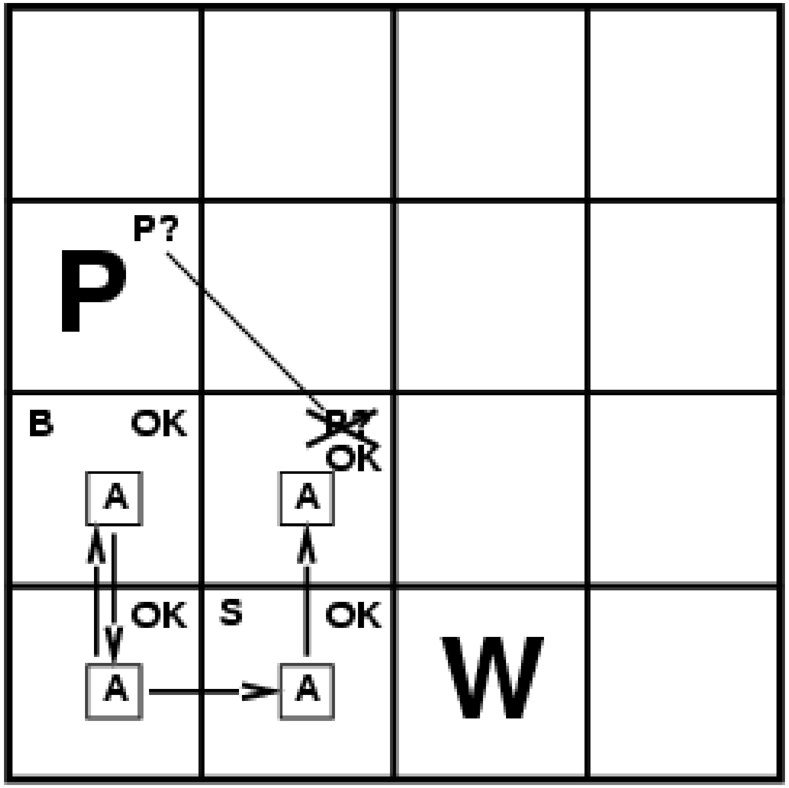
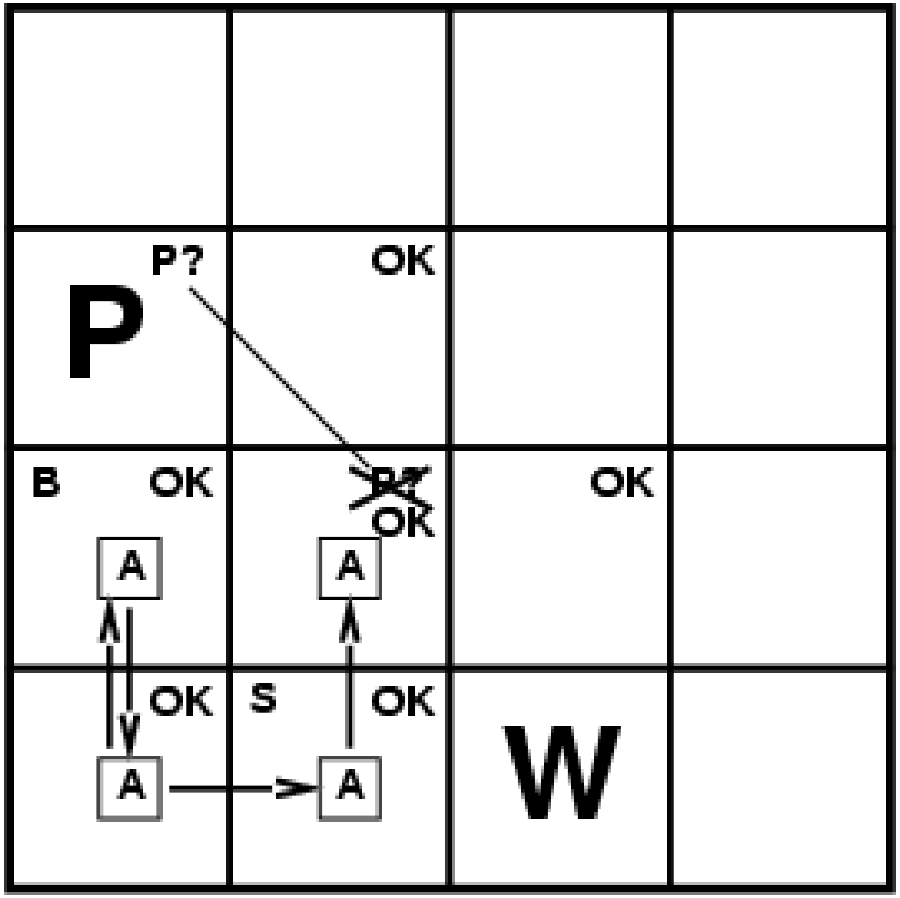


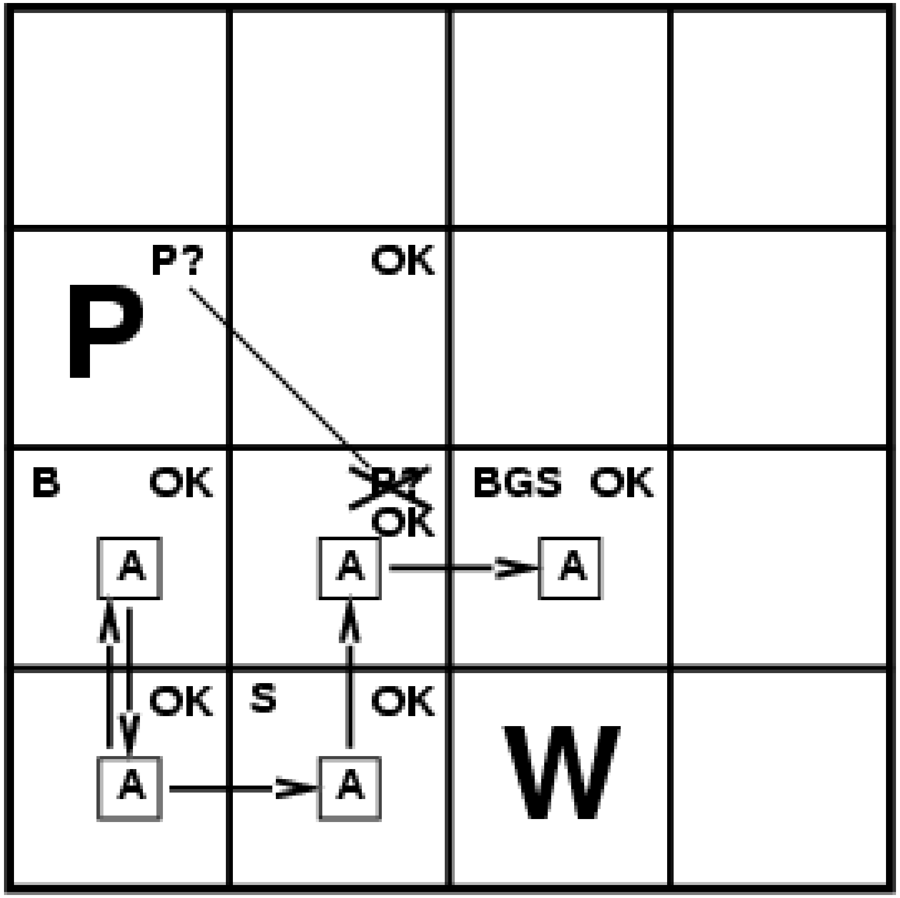
Another Wumpus World Game



Another Wumpus World Game



Another Wumpus World Game



Logic

- Logical Agents: combine the current percept with the existing knowledge (percept history) and do logical reasoning in identifying state information
- Reasoning conducted or assessed according to available knowledge about the domain
- **Logics** are formal languages for representing information such that conclusions can be drawn
- **Syntax**: Every sentence/premise in our knowledge base are expressed according to a syntax of the representing language.
- E.g., in Arithmetic, "x + y = 4" is a well-formed sentence, "xy+4=" is not

Logic – Semantics

- **Semantics**: Meaning of sentence. They define the truth of a sentence according to a possible world.
- E.g., in a world where $x=2$, $y=2$, "$x + y = 4$" is a true sentence, whereas in a world where $x=1$, $y=1$, the same sentence is not true
- In standard Logic, every sentence/premise has to be either True or False but cannot be in between

Logic – Model

- **Model:** Any possible world. Not necessarily the reality. Any combination of assignment of truth values to sentences in our KB.
- E.g., Two sentences S1 and S2 in our KB.
- Possible models: m1: {S1: True, S2: True}, m2: {S1: False, S2: True}, m3: {S1:True, S2: False}, m4: {S1:False, S2:False}
- S1 is true in m1, m3 → m1 satisfies S1 or m3 satisfies S1
- M(S1) is set of models that satisfy S1, i.e., [m1, m3]

Logic – Entailment

- Logical Reasoning: Logical entailment between sentences, i.e., a sentence follows from another sentence
- X $\models$ Y , it means that sentence X entails the sentence Y
- In every model where X is true, Y is also true
- Example: In Wumpus World, the agent is in [2,1] and detected a breeze
- The agent is interested in squares [1, 2], [2, 2], [3, 1] for next move.
- Now, each square might or might not contain a pit (total $2^3 = 8$ possible models)
- Our KB tells us that in [1, 1] we didn't receive a breeze and hence [2, 1] doesn't have a pit

Propositional Logic

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- A proposition is the basic building block of logic. It is defined as a declarative sentence that is either True or False, **but not both.** The area of logic which deals with propositions is called propositional logic
- The Truth Value of a proposition is True(denoted as T) if it is a true statement, and False(denoted as F) if it is a false statement.
  - The sun rises in the East and sets in the West. (Truth value = True)
  - 1 + 1 = 2 (Truth value = True)
  - 'b' is a vowel. (Truth value = False)
  - Will you go to office today? (No truth value therefore not a proposition)
  - X-3 = 6 (truth value can be True or False therefore not a proposition)
- Truth table is a tabular compilation of all the possible combinations of the propositions which are joined together by Logical Connectives to form the given compound proposition
- **Proposition Symbol** – A symbol that stands for a proposition.
  - E.g., $W_{1,3}$ – "Wumpus in [1,3]" is a proposition and $W_{1,3}$ is the symbol
  - Usually represented with uppercase letter and may contain other letters or subscripts. E.g., P, Q, R, $W_{1,3}$, North

Propositional Logic – Syntax

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- Syntax – Defines the language for allowable sentences
- Atomic Sentence – consist of a single propositional symbol
- Complex Sentence – Constructed from atomic sentences using parenthesis and logical connectives
- Logical connectives. There are five connectives
  - $\neg$ (not) – A sentence like $\neg W_{1,3}$ is the negation of $W_{1,3}$. A literal is either an atomic sentence (positive literal) or a negated atomic sentence (negative literal)
  - $\wedge$ (and) – Used in AND combination of sentences, i.e., $W_{1,3} \wedge P_{3,1}$. It is called a **conjunction** and its parts are called **conjuncts**
  - $\vee$ (or) – Used in OR combination of sentences, i.e., $W_{1,3} \vee P_{3,1}$. It is called a **disjunction** and its parts are called **disjuncts**
  - $\Rightarrow$ (implies) – A sentence such as $(W_{1,3} \wedge P_{3,1}) \Rightarrow \neg W_{2,2}$ is called an **implication** (or conditional). Its premise (antecedent) is LHS and its conclusion (consequent) is RHS
  - $\Leftrightarrow$ (if and only if) – A sentence like $W_{1,3} \Leftrightarrow \neg W_{2,2}$ is a **biconditional**

Propositional Logic - Negation

- Negation – If p is a proposition, then the negation of p is denoted by ¬p, which when translated to simple English means-
- “It is not the case that p” or simply “not p”.
- The truth value of ¬p is the opposite of the truth value of p.

p	$\neg p$
T	F
F	T

Propositional Logic - Conjunction

- Conjunction – For any two propositions p and q, their conjunction is denoted by p ∧ q, which means “p and q”.
- The conjunction p ∧ q is True when both p and q are True, otherwise False.
- The truth table of p ∧ q is:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Propositional Logic - Disjunction

- Disjunction – For any two propositions p and q, their disjunction is denoted by p V q, which means “p or q”.
- The disjunction p V q is True when either p or q is True, otherwise False.
- The truth table of p V q is:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Propositional Logic – Implication

- Implication – For any two propositions p and q, the statement "if p then q" is called an implication and it is denoted by $p \Longrightarrow q$.
- In the implication $p \Longrightarrow q$, p is called the hypothesis or antecedent or premise and q is called the conclusion or consequence.
- The implication is $p \Longrightarrow q$ is also called a conditional statement.
- The implication is false when p is true and q is false otherwise it is true. The truth table of $p \Longrightarrow q$ is:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$p \Longrightarrow q$ is true when p is false because the implication guarantees that when p and q are true then the implication is true. But the implication does not guarantee anything when the premise p is false. There is no way of knowing whether or not the implication is false since p did not happen.

Propositional Logic - Biconditional

- Biconditional or Double Implication – For any two propositions p and q, the statement "p if and only if(iff) q" is called a biconditional and it is denoted by $p \Leftrightarrow q$.
- The statement $p \Leftrightarrow q$ is also called a bi-implication.
- $p \Leftrightarrow q$ has the same truth value as $(p \Longrightarrow q) \land (q \Longrightarrow p)$
- The implication is true when p and q have same truth values, and is false otherwise. The truth table of $p \Leftrightarrow q$ is:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Propositional Logic – Syntax

- Operator Precedence: $\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$
- In ambiguous sentences like $\neg\,\mathrm{A}\ \wedge B$, resolution is done using precedence, such that $(\neg\,\mathrm{A})\ \wedge B$ rather than $\neg\,(\mathrm{A}\ \wedge B)$

Representation by Propositional Logic

For each [x, y] location

$P_{x,y}$ is true if there is a pit in [x, y]

$W_{x,y}$ is true if there is a wumpus in [x, y]

$B_{x,y}$ is true if agent perceives a breeze in [x, y]

$S_{x,y}$ is true if agent perceives a stench in [x, y]

--------- R is the sentence in KB--------------

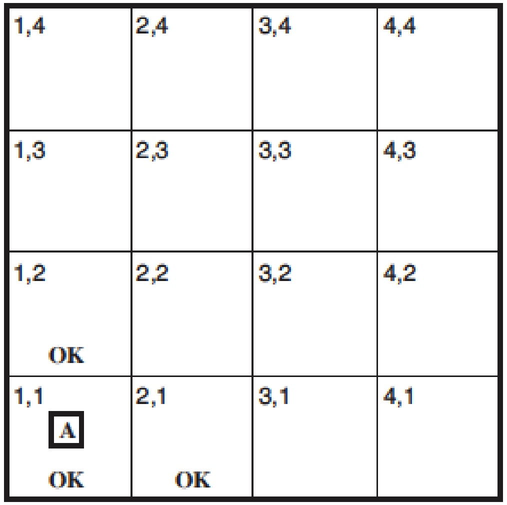
$R_1 : \neg P_{1,1}$

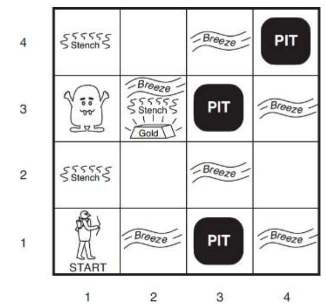
$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \lor P_{2,1})$

$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \lor P_{2,2} \lor P_{3,1})$

$R_4 : \neg B_{1,1}$

$R_5 : B_{2,1}$





Query **: $\neg$ $P_{1,2}$ entailed by our KB?**

Truth Table (TT) Entails Inference Example

$\neg P_{1,2}$ entailed by our KB?

1. Get sufficient information $B_{1,1}$, $B_{2,1}$, $P_{1,1}$, $P_{1,2}$, $P_{2,1}$, $P_{2,2}$, $P_{3,1}$
2. Enumerate all models with combination of truth values to propositional
symbols
3. In all the models, find those models where KB is true, i.e., every sentence R_1, R_2, R_3, R_4, R_5 are true
4. In those models where KB is true, find if query sentence $\neg P_{1,2}$ is true
5. If query sentence $\neg P_{1,2}$ is true in all models where KB is true, then it entails, otherwise it won't

Truth Table (TT) Entails Inference Example

$\neg$ $P_{1,2}$ entailed by our KB?

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	*false*	*false*	*false*	*false*	*false*	*false*	*true*	*true*	*true*	*true*	*false*	*false*
false	*false*	*false*	*false*	*false*	*false*	*true*	*true*	*true*	*false*	*true*	*false*	*false*
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
false	*true*	*false*	*false*	*false*	*false*	*false*	*true*	*true*	*false*	*true*	*true*	*false*
false	*true*	*false*	*false*	*false*	*false*	*true*	*true*	*true*	*true*	*true*	*true*	*true*
false	*true*	*false*	*false*	*false*	*true*	*false*	*true*	*true*	*true*	*true*	*true*	*true*
false	*true*	*false*	*false*	*false*	*true*	*true*	*true*	*true*	*true*	*true*	*true*	*true*
false	*true*	*false*	*false*	*true*	*false*	*false*	*true*	*false*	*false*	*true*	*true*	*false*
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
true	*true*	*true*	*true*	*true*	*true*	*true*	*false*	*true*	*true*	*false*	*true*	*false*

Truth Table (TT) Entails Inference Example

- In three models, KB is true
  - $\neg P_{1,2}$ is true in all three of those models,
  - hence, $\neg P_{1,2}$ is entailed from KB, i.e., $\neg P_{1,2}$ is true
  - There is no pit in [1, 2]
  - $P_{2,2}$ is true in two of the three models and false in one, so we cannot infer if there is a pit in [2,2] based on our simple KB

Model checking algorithm

- If KB and α contain n symbols, then there will be 2^n models
- Time Complexity: $O(2^n)$
- Space Complexity: O(n) because enumeration is depth-first

Inference : Example – Theorem Proving (Self Study)

- Theorem Proving – applying rules of inference directly to sentences in our KB to prove query sentence without consulting models
- **Logical Equivalence**: two sentences α and β are logically equivalent if they are true in the same set of models denoted as

$$\alpha \equiv \beta$$

E.g., $(P_{1,2} \vee B_{2,1})$ and $(B_{2,1} \vee P_{1,2})$ are logically equivalent

Inference : Example – Theorem Proving (Self Study)

- Logical Equivalence

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha \quad \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) \quad \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) \quad \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) \quad \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Inference : Example – Theorem Proving (Self Study)

Inference Rules

- Modus Ponens (Latin for Mode that affirms) : $\dfrac{\alpha \Rightarrow \beta, \alpha}{\beta}$
  - Whenever any sentence of the form $(\alpha \Rightarrow \beta)$ and α are given, then β can be inferred
  - Statement 1: If I am tired, then I need to rest. => $\alpha \rightarrow \beta$
  - Statement 2: I am tired => α
  - Conclusion: Therefore, I need to rest => β
- And-Elimination: Given a conjecture, any of the conjuncts can be inferred: $(\dfrac{\alpha \wedge \beta}{\alpha})$ or $(\dfrac{\alpha \wedge \beta}{\beta})$
  - E.g., If *(Tired* ∧ *Sleepy)* is given, then *Tired* can be inferred. Sleepy can also be inferred.

Inference using Theorem Proving - Example

$R_1 : \neg P_{1,1}$

$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

$R_4 : \neg B_{1,1}$

$R_5 : B_{2,1}$

Query: $\neg P_{1,2}$. Can we prove if this premise be entailed from KB using inference rules?

$$
\begin{array}{rcll}
(\alpha \wedge \beta) & \equiv & (\beta \wedge \alpha) & \text{commutativity of } \wedge \\
(\alpha \vee \beta) & \equiv & (\beta \vee \alpha) & \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) & \equiv & (\alpha \wedge (\beta \wedge \gamma)) & \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) & \equiv & (\alpha \vee (\beta \vee \gamma)) & \text{associativity of } \vee \\
\neg(\neg \alpha) & \equiv & \alpha & \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) & \equiv & (\neg \beta \Rightarrow \neg \alpha) & \text{contraposition} \\
(\alpha \Rightarrow \beta) & \equiv & (\neg \alpha \vee \beta) & \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) & \equiv & ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) & \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) & \equiv & (\neg \alpha \vee \neg \beta) & \text{De Morgan} \\
\neg(\alpha \vee \beta) & \equiv & (\neg \alpha \wedge \neg \beta) & \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) & \equiv & ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) & \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) & \equiv & ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) & \text{distributivity of } \vee \text{ over } \wedge
\end{array}
$$

Inference using Theorem Proving - Example

$R_1 : \neg P_{1,1}$

$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

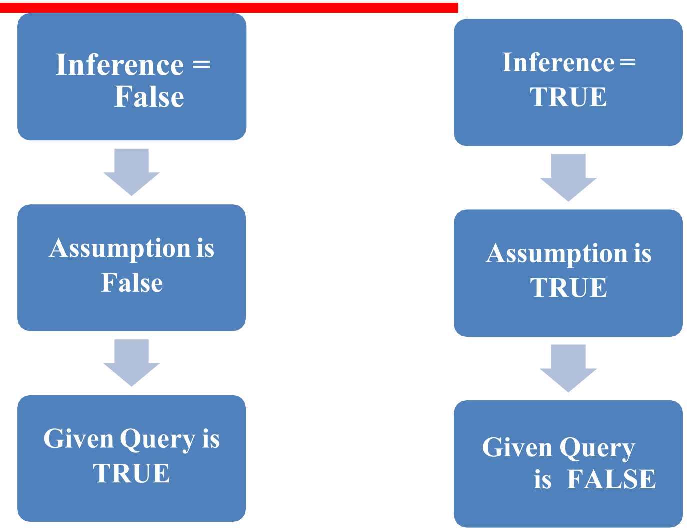
$R_4 : \neg B_{1,1}$

$R_5 : B_{2,1}$

Query: $\neg P_{1,2}$. Can we prove if this premise be entailed from KB using inference rules?

$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$	
$R_6 : (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$	Biconditional Elimination
$R_7 : ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$	And Elimination
$R_8 : (\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$	Contraposition
$R_9 : \neg (P_{1,2} \vee P_{2,1})$	Modus Ponens
$R_{10} : \neg P_{1,2} \wedge \neg P_{2,1}$	Demorgans
R11: $\neg P_{1,2}$	And Elimination

Propositional Logic – Proof By Contradiction



PL Resolution Using CNF Conversion

$R_1 : \neg P_{1,1}$

$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \lor P_{2,1})$

$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \lor P_{2,2} \lor P_{3,1})$

$R_4 : \neg B_{1,1}$

$R_5 : B_{2,1}$

Query: $\neg P_{1,2}$

The conjunctive normal form states that a formula is in CNF if it is a conjunction of one or more than one clause, where each clause is a disjunction of literals. In other words, it is a product of sums where $\wedge$ symbol occurs between the clauses and the $\vee$ symbol occurs in the clauses.

CNF:
$(A \vee B) \wedge (A \vee B \vee \sim C) \wedge \sim A$
Unit resolution: $\sim A$
Query: $\sim C$

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$R_5 : B_{2,1}$

Query: $\neg P_{1,2}$

PL-Resolution

$(\alpha \land \beta) \equiv (\beta \land \alpha)$ commutativity of $\land$
$(\alpha \lor \beta) \equiv (\beta \lor \alpha)$ commutativity of $\lor$
$((\alpha \land \beta) \land \gamma) \equiv (\alpha \land (\beta \land \gamma))$ associativity of $\land$
$((\alpha \lor \beta) \lor \gamma) \equiv (\alpha \lor (\beta \lor \gamma))$ associativity of $\lor$
$\neg(\neg\alpha) \equiv \alpha$ double-negation elimination
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$ contraposition
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \lor \beta)$ implication elimination
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \land (\beta \Rightarrow \alpha))$ biconditional elimination
$\neg(\alpha \land \beta) \equiv (\neg\alpha \lor \neg\beta)$ De Morgan
$\neg(\alpha \lor \beta) \equiv (\neg\alpha \land \neg\beta)$ De Morgan
$(\alpha \land (\beta \lor \gamma)) \equiv ((\alpha \land \beta) \lor (\alpha \land \gamma))$ distributivity of $\land$ over $\lor$
$(\alpha \lor (\beta \land \gamma)) \equiv ((\alpha \lor \beta) \land (\alpha \lor \gamma))$ distributivity of $\lor$ over $\land$

$R_1 : \neg P_{1,1}$

~~$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \lor P_{2,1})$~~

~~**$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \lor P_{2,2} \lor P_{3,1})$**~~

$R_4 : \neg B_{1,1}$

$R_5 : B_{2,1}$

Query: $\neg P_{1,2}$

Eliminate		**$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \lor P_{2,1})$**	**$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \lor P_{2,2} \lor P_{3,1})$**
$\leftarrow\rightarrow$	Biconditional Elimination		
$\rightarrow$	Implication Elimination		
Clause level $\neg$	De Morgan		
CNF Form	Distributivity of $\lor$ over $\land$		

PL-Resolution

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$ commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$ associativity of $\wedge$
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$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$ distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$ distributivity of $\vee$ over $\wedge$

$R_1 : \neg P_{1,1}$
$\cancel{R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})}$
$\cancel{\mathbf{R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})}}$
$R_4 : \neg B_{1,1}$
$R_5 : B_{2,1}$
Query: $\neg P_{1,2}$

Eliminate		$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$	$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
←→	Biconditional Elimination	$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$	$(B_{2,1} \Rightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})) \wedge ((P_{1,1} \vee P_{2,2} \vee P_{3,1}) \Rightarrow B_{2,1})$
→	Implication Elimination	$\neg B_{1,1} \vee (P_{1,2} \vee P_{2,1})$ $\mathbf{\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1}}$	$\neg B_{2,1} \vee (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ $\mathbf{\neg(P_{1,1} \vee P_{2,2} \vee P_{3,1}) \vee B_{2,1}}$
Clause (level ¬	De Morgan	$(\neg \mathbf{P_{1,2}} \wedge \neg \mathbf{P_{2,1}}) \vee \mathbf{B_{1,1}}$	$(\neg \mathbf{P_{1,1}} \wedge \neg \mathbf{P_{2,2}} \wedge \neg \mathbf{P_{3,1}}) \vee \mathbf{B_{2,1}}$
CNF Form	Distributivity of V over Λ	$(\neg \mathbf{P_{1,2}} \vee \mathbf{B_{1,1}}) \wedge (\neg \mathbf{P_{2,1}} \vee \mathbf{B_{1,1}})$	$(\neg \mathbf{P_{1,1}} \vee \mathbf{B_{2,1}}) \wedge (\neg \mathbf{P_{2,2}} \vee \mathbf{B_{2,1}}) \wedge (\neg \mathbf{P_{3,1}} \vee \mathbf{B_{2,1}})$

PL-Resolution

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$ commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$ associativity of $\wedge$
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$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$ distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$ distributivity of $\vee$ over $\wedge$

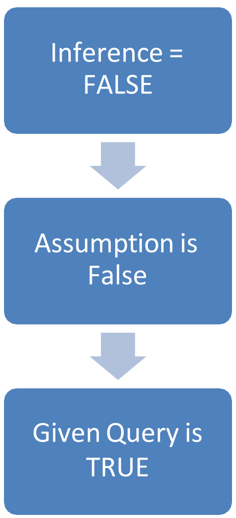
$R_1 : \neg P_{1,1}$
$\cancel{R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})}$
$\cancel{\mathbf{R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})}}$
$R_4 : \neg B_{1,1}$
$R_5 : B_{2,1}$
Query: $\neg P_{1,2}$

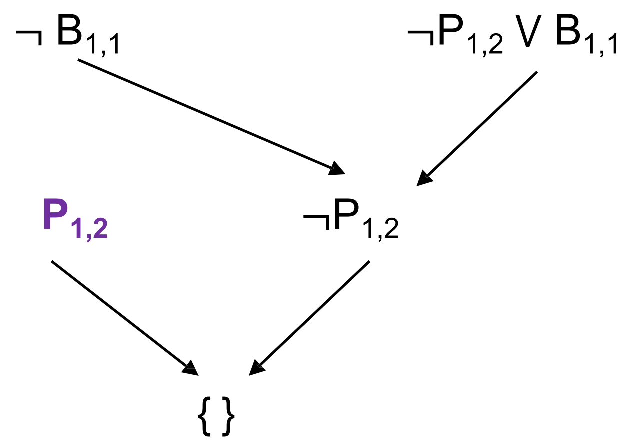
$R_6 : \neg B_{1,1} \vee P_{1,2} \vee P_{2,1}$
$R_7 : \neg P_{1,2} \vee B_{1,1}$
$R_8 : \neg P_{2,1} \vee B_{1,1}$
$R_9 : \neg B_{2,1} \vee P_{1,1} \vee P_{2,2} \vee P_{3,1}$
$R_{10} : \neg P_{1,1} \vee B_{2,1}$
$R_{11} : \neg P_{2,2} \vee B_{2,1}$
$R_{12} : \neg P_{3,1} \vee B_{2,1}$

Eliminate		$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$	$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
←→	Biconditional Elimination	$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$	$(B_{2,1} \Rightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})) \wedge ((P_{1,1} \vee P_{2,2} \vee P_{3,1}) \Rightarrow B_{2,1})$
→	Implication Elimination	$\neg B_{1,1} \vee (P_{1,2} \vee P_{2,1})$ $\mathbf{\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1}}$	$\neg B_{2,1} \vee (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ $\mathbf{\neg(P_{1,1} \vee P_{2,2} \vee P_{3,1}) \vee B_{2,1}}$
Clause level ¬	De Morgan	$\mathbf{(\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1}}$	$\mathbf{(\neg P_{1,1} \wedge \neg P_{2,2} \wedge \neg P_{3,1}) \vee B_{2,1}}$
CNF Form	Distributivity of ∨ over ∧	$\mathbf{(\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})}$	$\mathbf{(\neg P_{1,1} \vee B_{2,1}) \wedge (\neg P_{2,2} \vee B_{2,1}) \wedge (\neg P_{3,1} \vee B_{2,1})}$

PL-Resolution

To find: Is there a pit in location (1,2) using the CNF obtained in previous slide. i.e.
Query: ~ $P_{1,2}$





$R_1 : \neg P_{1,1}$

$\cancel{R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \lor P_{2,1})}$

$\cancel{\mathbf{R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \lor P_{2,2} \lor P_{3,1})}}$

$R_4 : \neg B_{1,1}$

$R_5 : B_{2,1}$

Query: $\neg P_{1,2}$

$R_6 : \neg B_{1,1} \lor P_{1,2} \lor P_{2,1}$

$R_7 : \neg P_{1,2} \lor B_{1,1}$

$R_8 : \neg P_{2,1} \lor B_{1,1}$

$R_9 : \neg B_{2,1} \lor P_{1,1} \lor P_{2,2} \lor P_{3,1}$

$R_{10} : \neg P_{1,1} \lor B_{2,1}$

$R_{11} : \neg P_{2,2} \lor B_{2,1}$

$R_{12} : \neg P_{3,1} \lor B_{2,1}$

DPLL Algorithm

In logic and computer science, the Davis–Putnam–Logemann–Loveland (**DPLL**) **algorithm** is a complete, backtracking-based search **algorithm** for deciding the satisfiability of propositional logic formulae in conjunctive normal form

Improvements:

1. Early Termination
2. Pure Symbolic Heuristic
3. Unit Clause Heuristic

DPLL Algorithm

$R_7 : \neg P_{1,2} \vee B_{1,1}$
$R_8 : \neg P_{2,1} \vee B_{1,1}$
$\{P_{1,2}, B_{1,1}, P_{2,1}\}$

$R_7 : \neg P_{1,2} \vee B_{1,1}$
$R_8 : \neg P_{2,1} \vee B_{1,1}$

$R_7 : \neg P_{1,2} \vee B_{1,1}$
$R_8 : \neg P_{2,1} \vee B_{1,1}$

$R_7 : \neg P_{1,2} \vee B_{1,1}$
$R_8 : \neg P_{2,1} \vee B_{1,1}$

Sample Problem

"In the context of online shopping recommendations, if a customer frequently purchases electronics items, they are considered tech-savvy. Not all tech-savvy customers make expensive purchases, but all expensive purchases are made by tech-savvy customers. Tech-savvy customers always follow the latest trends in technology. Customers who follow tech trends will receive recommendations for new electronic products."

- Use propositional logic (without quantifiers) to efficiently represent the knowledge base given above.
- W.r.t the knowledge base from part (a), convert it into Conjunctive Normal Form (CNF) and find one sample complete Binary Satisfiability (BSAT) solution to the variables using the DPLL algorithm.
- Suppose a customer is known to frequently purchase electronics but has ruled out being tech-savvy (¬T) with 100% certainty. Under this scenario, if the logic agent is asked, "Is the customer likely to receive recommendations for new electronic products?" how does the agent reason to answer the query? Prove by contradiction using the results obtained in part (a) by the resolution method.

Pros & Cons of Propositional Logic

innovate achieve lead

- ☺ Propositional logic is declarative
- ☺ Propositional logic allows partial/disjunctive/negated information
  - (unlike most data structures and databases
- ☺ Propositional logic is compositional:
  - meaning of $B_{1,1} \wedge P_{1,2}$ is derived from meaning of $B_{1,1}$ and of $P_{1,2}$
- ☺ Meaning in propositional logic is context-independent
  - (unlike natural language, where meaning depends on context)
- ☹ Propositional logic has very limited expressive power
  - (unlike natural language)
  - E.g., cannot say "pits cause breezes in adjacent squares“
    - except by writing one sentence for each square
  - Ex: Some students are brilliant.

Citations and References

- Few content for these slides may have been obtained from prescribed books and various other source on the Internet
- I hereby acknowledge all the contributors for their material and inputs and gratefully acknowledge people others who made their course materials freely available online.
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Thank You for your time & attention !

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